MANJITHA D VIDANALAGE

Data Privacy & AI Researcher

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Date of birth	30.11.1991 in Sri Lanka
Nationality	Sri Lankan
Residence Status	EU Blue Card
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EDUCATION

M. Sc. ELECTRICAL ENGINEERING AND INFORMATION TECHNOLOGY *[Oct 2021 – Oct 2023]* Final Grade : 1.6
Deggendorf Institute of Technology, Germany

POST GRADUATE DIPLOMA in ELECTRICAL INSTALLATIONS *[Sep 2017 – Oct 2018]*
B.Sc. ENG (HONS.) ELECTRICAL ENGINEERING *[Sep 2011 – Apr 2016]* Final Grade : 1.9
University of Moratuwa , Sri Lanka

WORK EXPERIENCE

RESEARCH ASSOCIATE (Wissenschaftlicher Mitarbeiter) *[Mar 2024 – Present]*
Deggendorf Institute of Technology, Technology Campus Vilshofen, Germany

- Conducting in-depth research on AI applications in data privacy, anonymization, and open data systems, staying at the forefront of emerging AI trends and innovations
- Evaluating privacy-preserving AI technologies and methodologies, providing technical insights for the EAsyAnon research project valued at substantial research funding
- Leading development of privacy risk audit systems integrating advanced AI agents (LLMs, RAG, LangGraph) with mathematical algorithms for comprehensive data evaluation
- Supervising working students, evaluating technical capabilities, and fostering research excellence in AI and data science
- Contributing to grant proposal development, presenting research findings at academic conferences, and building networks within the AI research community
- Translating complex AI and privacy concepts for diverse audiences including academic peers, industry partners, and funding bodies

MASTER'S THESIS STUDENT *[Mar 2023 – Nov 2023]*
Deggendorf Institute of Technology, Technology Campus Vilshofen, Germany

- Developed the mathematical framework for the audit system for anonymised tabular data for the project EAsyAnon
  Thesis : Scoring System for Quantifying the Privacy in Re-Identification of Tabular Datasets *(Grade : 1.2)*

STUDENT ASSISTANT *[Apr 2022 – Feb 2023]*
Deggendorf Institute of Technology, Technology Campus Teisnach Sensorik , Germany

- Executing projects in automation, electronics and networking and meeting the project goals and timelines.

HEAD OF DEPARTMENT [Design] / ELECTRICAL ENGINEER *[Nov 2019 – Nov 2021] / [Jan 2017 – Nov 2019]*
Amithi Power Consultants (Private) Limited, Rajagiriya, Sri Lanka

- Led engineering projects managing teams and ensuring timely completion
- Designed and tested high-voltage power distribution systems with protection and control schemes

ELECTRICAL ENGINEER *[May 2016 – Jan 2017]*
Building Services Consultants (Private) Limited, Nawala, Sri Lanka

- Handling projects in Lighting, Fire protection & HVAC adhering to international standards such as IEC, BS & NFPA ensuring on-time completion and within budget

PROJECTS

EAsyAnon *https://www.easyanon.de/*

AutoAudit : System for calculating re-identification risk of tabular datasets *[Mar 2023 - Present]*

- **Developing the mathematical framework**
  Technologies used : Python with NumPy, Pandas, Matplotlib
  - Built an automated scoring system to assess re-identification risks in open datasets, implementing conventional metrics (k-anonymity, l-diversity, t-closeness) and novel risk metrics for comprehensive privacy evaluation
  - Evaluating and validating the system across multiple real-world tabular datasets
- **Implementation of desktop application**
  Technologies used : Python Flask, Electron, JS, HTML /CSS, Django, PostgreSQL, Azure BLOB storage
  - Integration of the mathematical framework in to a desktop application with a Python Flask backend and an Electron frontend
  - Developed Django RESTful API, Azure BLOB Storage and PostgreSQL DB to upload the dataset and update data
- **AI agents integration**
  Technologies used : LLMs (GPT-4, Gemini-2.5), LangGraph
  - Integrations of Agentic AI pipeline in parallel with the mathematical algorithms to evaluate the privacy risk of semantics in text or string data

Privacy-Preserving Electricity Load Forecasting in Smart Cities Using TinyML at the Edge

Technologies used : TensorFlow Lite, Pyfhel

- Collaborated to develop TinyML models for on-device electricity consumption forecasting
- Collaborated to develop a homomorphic encryption scheme for secure data aggregation
- Presented the research paper in SmartCity-2025 conference in Exeter, UK

SKILLED TECHNOLOGIES

AI & ML	: TensorFlow, PyTorch, Scikit-learn, LLMs, RAG, LangGraph
Programming & Data Science	: Python, NumPy, Pandas, Matplotlib
Cloud & Infrastructure	: Azure, PostgreSQL, Azure BLOB Storage
Development Tools	: Flask, Django, Electron, Git, HTML/CSS, JavaScript
Specialized Tools	: ARX Data Anonymisation Tool, TensorFlow Lite, Pyfhel, LaTeX

PUBLICATIONS

1. Almaini, A., Nataraj, S., Savaliya, G., Vidanalage, M. D., Subedi, A., Al-Dubai, A., & Folz, J. Privacy-Preserving Electricity Load Forecasting in Smart Cities Using TinyML at the Edge. 23rd IEEE International Conference on Smart City (SmartCity-2025).
2. Folz, J., Vidanalage, M. D., Aufschläger, R., Almaini, A., Heigl, M., Fiala, D., & Schramm, M. (2025). Scoring System for Quantifying the Privacy in Re-Identification of Tabular Datasets. *IEEE Access*. DOI: 10.1109/ACCESS.2025.3563309.
3. Vidanalage, M. D., Folz, J., Heigl, M., Wilhelm, S., & Schramm, M. (2025). System for Calculating Open-data Re-identification Risk (SCORR). Zenodo. DOI : 10.5281/zenodo.16740751.
4. Folz, J., Guggumos, J., Vidanalage, M. D., Almaini, A., Fiala, D., Heigl, M., & Schramm, M. (2025). From anonymisation to encryption: A comprehensive evaluation of privacy-enhancing technologies for machine learning. To be presented at ICICT 2026, London, UK.

LANGUAGES

Sinhalese : Mother tongue **English :** Proficient (C1) **German :** Intermediate (B1)

INTERESTS

AI Agents Data Privacy NLP Privacy-Preserving ML Synthetic Data Generation

HOBBIES

Landscape, Travel & Architectural Photography

Passau, 26.10.2025